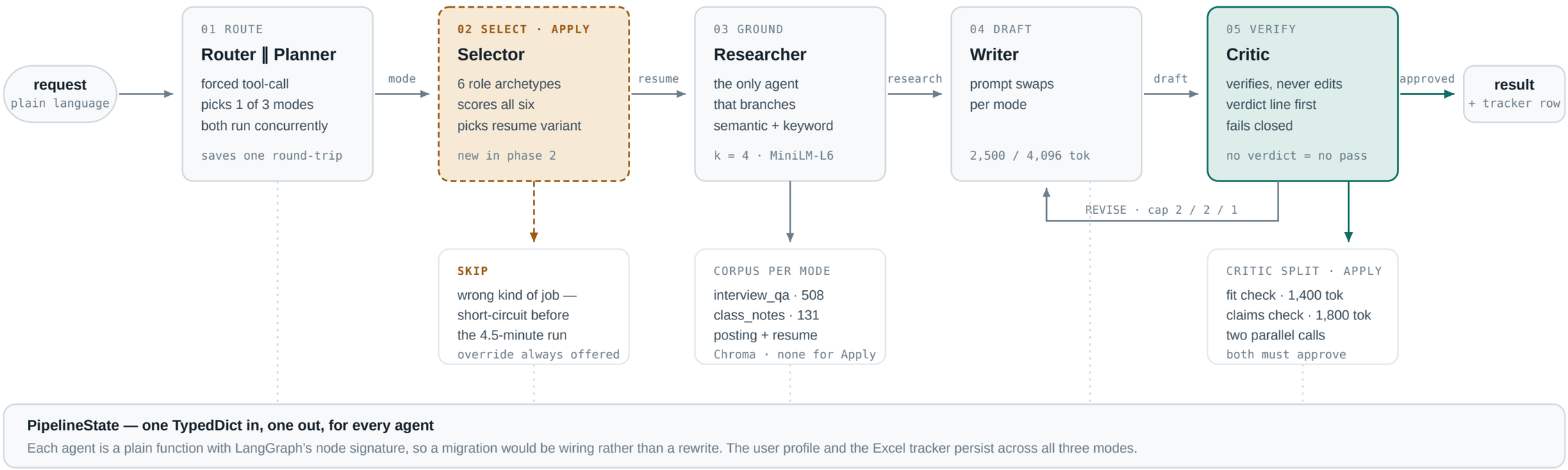
MULTI-AGENT CAPSTONE · UPDATED ARCHITECTURE

Career Copilot Architecture

Three modes, one pipeline. Apply tailors a resume to a posting and judges whether you qualify; Interview quizzes you; Learn revises a concept from your own class notes. Mode-specific behaviour lives only in system prompts and corpus choice — never in duplicated code.



The spine is identical in all three modes. Only two things are Apply-specific: the Selector gate at step 02, and the Critic splitting into two parallel calls at step 05.

What actually differs per mode

Everything below is a prompt or a constant. No branching code paths.

MODE	CORPUS	WRITER	REV	RUNTIME
Apply	posting + resume	2,500	2	~4.5 min
Interview	interview_qa	4,096	2	70–85 s
Learn	class_notes	4,096	1	1–2.5 min

The binding constraint

Groq free tier: `8,000` tokens per minute, `200,000` per day. Apply's Critic is the call that sits closest to the ceiling.

```
research      ~4,200
draft         ~2,000
prompt          ~300
output        1,800
─────────────────────
total         ~8,300  vs  8,000
```

There is no headroom to raise budgets. The only lever is making inputs or outputs smaller — which is why the Critic was split in two, and why the Selector sends rubrics without resume text.

Build status

- **Router, Planner, Researcher, Writer, Critic** — built and passing, 16 / 16 unit tests, ten logged bugs fixed.
- **Selector** — built and tested standalone; **not yet wired** into the orchestrator. Six rubrics, a 20-posting regression set, and a parser that can never fail into a false skip.
- **Archetype accuracy** — 100% top-1 on the regression set, but no-match detection catches only 2 of 4. Being fixed by adding an explicit "none of these" class.
- **Apply → Interview loop** — designed, not built. Every requirement marked NOT MET becomes an interview topic.

Stack: Groq `openai/gpt-oss-120b` · LangChain + Chroma + `all-MiniLM-L6-v2` embeddings · Streamlit UI · openpyxl tracker. Agents are plain functions over a shared `PipelineState` TypedDict; a LangGraph migration was deliberately deferred because the function signatures already match its node contract.